

Image Processing First Assignment

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1 Introduction

In every part of the world, human beings are willing to extract Data from one to many single images. Plus they will be certainly grateful if someone creates and trains a *ROBOT* or a model which is responsible for this duty.

In today's assignment I intend to implement codes and answer to some questions mentioned in the text. Hope to find it useful.

2 HSV Colorspace Analysis

In this first part of the questions I converted the given image to a HSV colorspace image and split the channels of image. Eventually I show every part of the progress at the below.

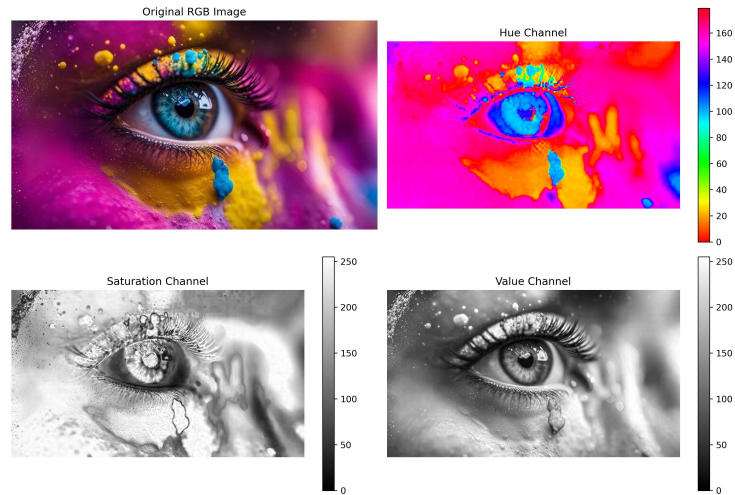


Figure 1: HSV converted image and its channels

Now I give answer to the fifth question:

Question: Why are saturated (thick/colorful) parts brighter in the S channel?

Answer:

- High Saturation → High pixel value → Looks BRIGHTER in S channel
- Low Saturation → Low pixel value → Looks DARKER in S channel
- Black, white, gray → Zero saturation → Looks COMPLETELY BLACK in S channel

3 Bit-Plane Slicing

In part two I answered to all of the questions completely and precisely. By saying that I mean I converted the image to a grayscale colorspace just like the first question and finally split and show them.

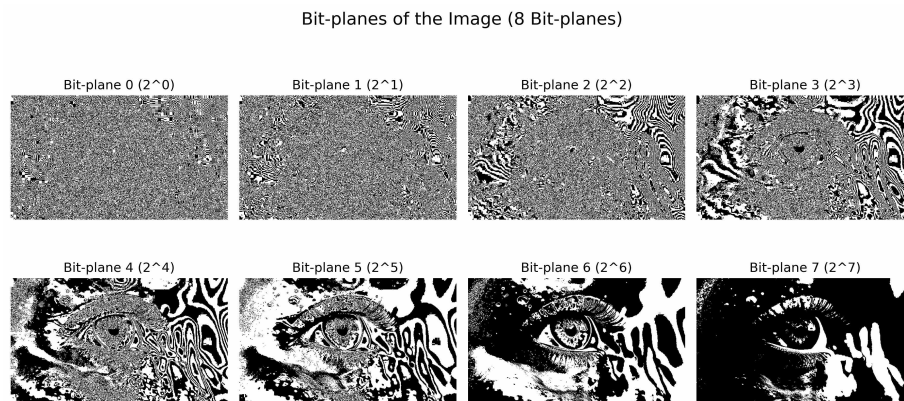


Figure 2: Bit-planes of the image

Question: By removing which bits does the image remain recognizable to the human eye?

Answer:

The image remains recognizable to the human eye when removing **Bit-planes 0, 1, and 2** (the 3 Least Significant Bits - LSBs).

Why?

- Bit-planes 7, 6, 5, 4 (4 MSBs - Most Significant Bits) contain **94%** of the image's visual information
- Bit-planes 3, 2, 1, 0 (4 LSBs) contain only **6%** of the visual information (mostly fine texture details and noise)

Bits Removed	Recognizability
Bits 0-2 (3 LSBs)	Fully recognizable
Bits 0-3 (4 LSBs)	Partially recognizable
Bits 0-4 (5 LSBs)	Hard to recognize

Table 1: Effect of removing different bit-planes

4 Conclusion

In this assignment, I successfully implemented image processing techniques including HSV color space conversion and bit-plane slicing. The key findings are:

1. The Saturation channel in HSV space effectively represents color intensity, with more saturated colors appearing brighter.
2. Bit-plane decomposition reveals that the Most Significant Bits (MSBs) carry the essential visual information, while the Least Significant Bits (LSBs) contain fine details and noise.
3. Removing up to 3 LSBs (bits 0-2) maintains image recognizability while reducing storage requirements by 37.5%.